

CAMM 535 Final Project — Report 1

Group 4: Schwannoma

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Abstract

This report documents data acquisition and the conceptual database design for Schwannoma. We integrate multiple biological databases to construct a unified schema linking phenotype-related genes, variants, genomic coordinates, proteins, and GEO2R results.

Contents

1	Part 1 — Disease / Phenotype Introduction	3
1.1	Clinical characteristics	3
1.1.1	Annual incidence (sporadic vestibular schwannoma)	3
1.2	Biological relevance	3
1.3	Known genetic associations	3
2	Part 2: Gene Selection and Variant Retrieval	6
2.1	Step 1: Identification of Schwannoma-Associated Genes Using BioMart	6
2.2	Step 2: Protein–Protein Interaction Expansion Using STRING	6
2.3	Step 3: Variant Retrieval Using Ensembl BioMart	8
2.4	Automated Variant Retrieval Script (Final Working Version)	9
3	Part 3: Genomic Coordinates and Transcript Annotation	17
3.1	Data Source and Genome Assembly	19
3.2	Input Identifiers	19
3.3	Selected Attributes	20
3.4	Transcript-Level Representation	20
3.5	Output Format and Downstream Use	20
4	Part 4 — UniProt ID Mapping (Genes → Proteins)	21
5	Part 5 — GEO Dataset & GEO2R	21

6	Part 6 — Additional Table from Another Database	21
6.1	Integration of TED and AlphaFold Database	22
6.2	The Truncated Evolutionary Domain (TED) Database	22
6.3	CATH Structural Classification	23
6.4	Integration with AlphaFold Database (AFDB)	23
6.5	Impact and Applications	23
7	Part 7 — Conceptual Design (ER Diagram + Keys)	23
7.1	Entity-Relationship diagram	23
7.2	Primary keys and foreign keys	23
8	Challenges and Ambiguities	24
8.1	Code Evolution and Rationale (Old vs. New Implementation)	24

1 Part 1 — Disease / Phenotype Introduction

Schwannoma is a typically benign, slow-growing peripheral nerve sheath tumor that arises from Schwann cells, the glial cells responsible for myelination of peripheral nerves. Schwannomas most commonly affect cranial and spinal nerves and are frequently associated with the vestibular branch of the eighth cranial nerve, where they are known as vestibular schwannomas (also called acoustic neuromas) [1].

1.1 Clinical characteristics

Clinically, schwannomas usually present as well-circumscribed, encapsulated tumors. Symptoms depend on tumor location and size and may include hearing loss, tinnitus, balance disturbances, localized pain, or neurological deficits due to nerve compression. Vestibular schwannomas represent the most common subtype and account for approximately 8–10% of all intracranial tumors [2]. Although most schwannomas are sporadic, bilateral vestibular schwannomas are a hallmark feature of neurofibromatosis type 2 (NF2), a hereditary tumor predisposition syndrome [3].

1.1.1 Annual incidence (sporadic vestibular schwannoma)

Population-based registry data suggest that sporadic vestibular schwannoma (VS) is diagnosed at an annual rate on the order of a few cases per 100,000 people. In a recent UK cohort registry study (2013–2016), the mean annual incidence of newly diagnosed VS was reported as **2.2 per 100,000 person-years**, with incidence rising strongly with age and peaking in the **60–69** year group (about **5.8 per 100,000 person-years**) [8]. Consistent with this scale, recent US population-based surveillance (CBTRUS, 2017–2021) reports an age-adjusted incidence rate for **vestibular schwannoma of 1.52 per 100,000** and shows that vestibular schwannoma constitutes the majority of non-malignant nerve sheath tumors in the CNS (Figure 1) [9].

1.2 Biological relevance

From a biological perspective, schwannomas provide an important model for studying tumor suppressor gene dysfunction and dysregulated cell signaling in glial-derived tumors. Schwann cells play a critical role in axonal support and nerve regeneration, and disruption of their growth control mechanisms can lead to uncontrolled proliferation. Schwannoma cells typically retain a benign histological appearance, yet they can cause significant morbidity due to their anatomical location and compressive growth pattern [4].

1.3 Known genetic associations

...

The most well-established genetic association in schwannoma is loss-of-function mutation or deletion of the *NF2* gene, which encodes the tumor suppressor protein merlin (schwannomin). Merlin is a key regulator of contact inhibition and multiple signaling pathways, including Hippo, PI3K/AKT, and MAPK pathways [5]. In sporadic schwannomas, somatic alterations of *NF2*

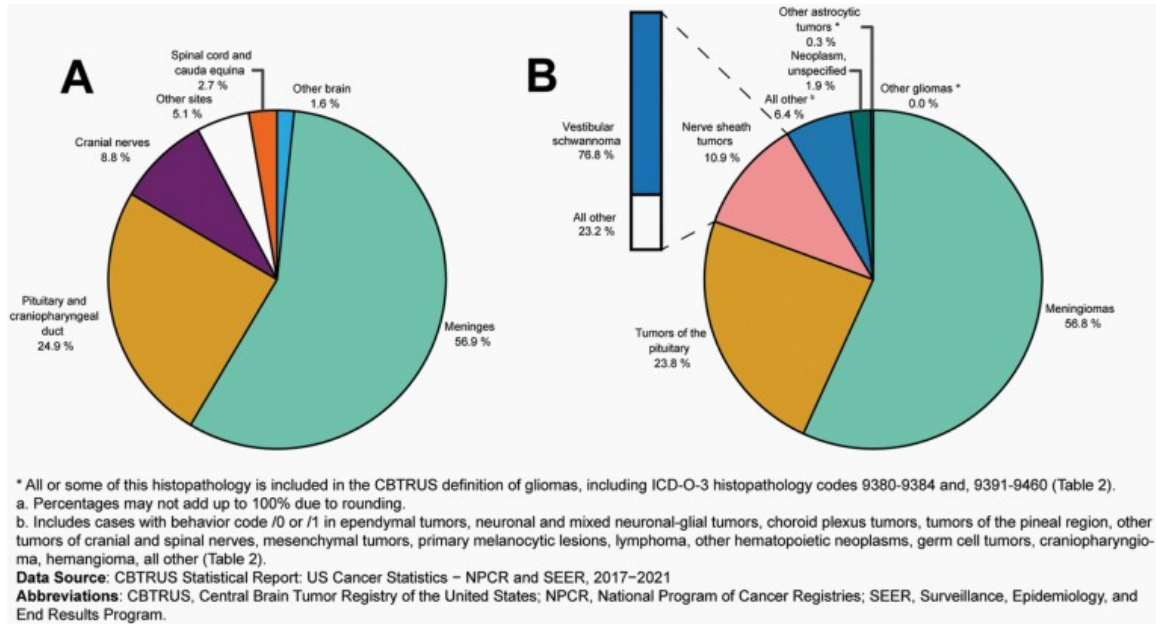


Figure 1. CBTRUS (US, 2017–2021): distribution of non-malignant primary brain/CNS tumors by (A) site and (B) histopathology. Panel B highlights that vestibular schwannoma represents a large fraction of nerve sheath tumors. Adapted from [9].

are observed in a majority of cases, while germline mutations are characteristic of patients with NF2-associated disease. Additional genetic and epigenetic alterations involving pathways related to cytoskeletal organization and cell adhesion have also been reported, highlighting the molecular heterogeneity of schwannomas [6].

As illustrated in Figure 2, disruption of merlin-mediated signaling removes critical growth-inhibitory signals in Schwann cells.

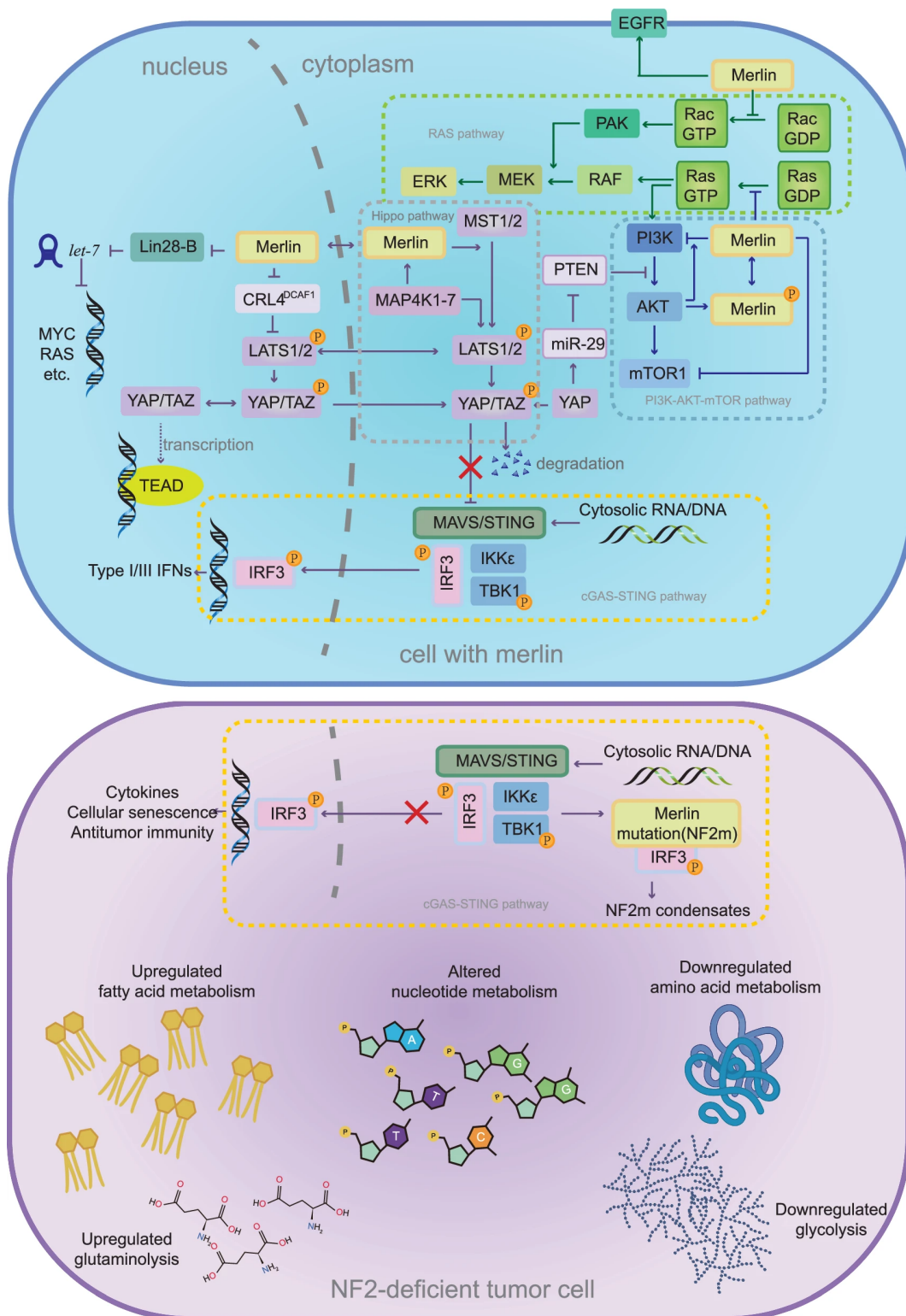


Figure 2. NF2/merlin signaling pathways in schwannoma development. Merlin regulates multiple growth-control pathways, including Hippo, PI3K/AKT, and MAPK signaling. Loss of NF2 function removes growth inhibition in Schwann cells and promotes tumor formation. Adapted from [7].

2 Part 2: Gene Selection and Variant Retrieval

2.1 Step 1: Identification of Schwannoma-Associated Genes Using BioMart

The initial set of genes associated with schwannoma was obtained using the Ensembl BioMart database. The *Human genes (GRCh38.p14)* dataset was selected, and the phenotype filter was set to *schwannoma*. From this dataset, core gene-related attributes were retrieved, including Ensembl Gene Stable ID, gene name, chromosome, and genomic coordinates.

This step ensured that the starting gene list was phenotype-driven and directly linked to schwannoma, rather than relying on manually curated or literature-derived gene sets.

Screenshots illustrating the BioMart gene selection and attribute configuration are provided in Figure 3.

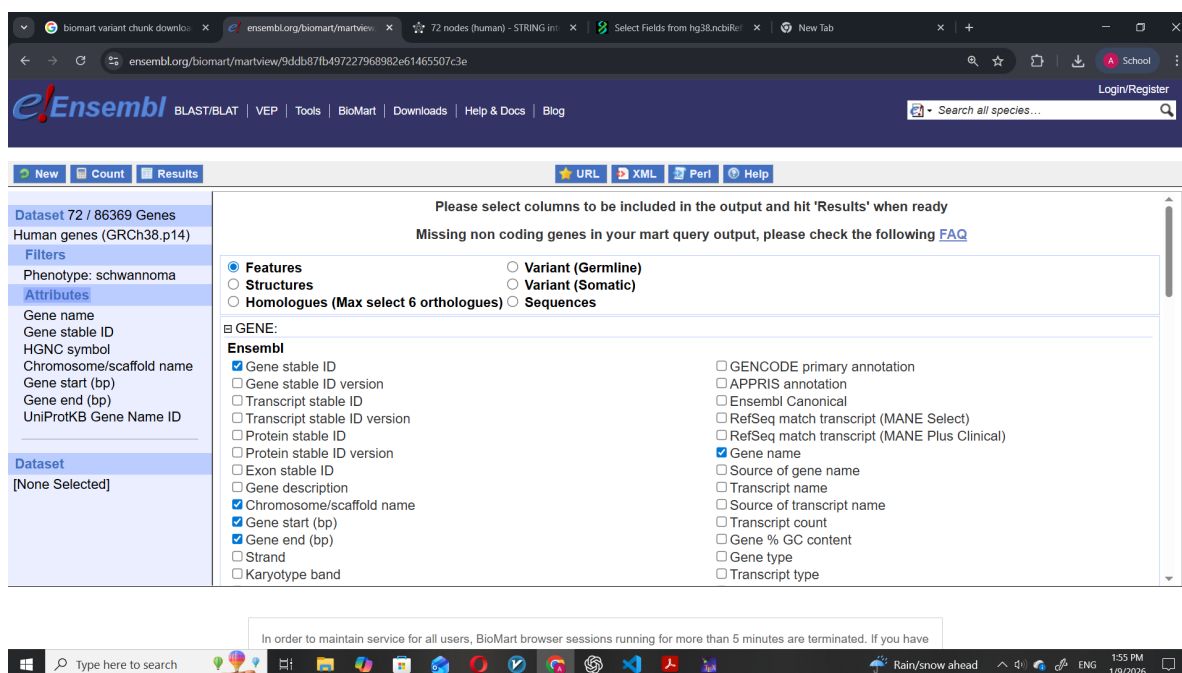


Figure 3. BioMart interface showing the selection of schwannoma-associated genes and genomic attributes.

2.2 Step 2: Protein–Protein Interaction Expansion Using STRING

The gene list obtained from BioMart was subsequently used as input for the STRING database to analyze protein–protein interactions and expand the gene set based on biological connectivity. Gene symbols were entered using the *Multiple Proteins by Names / Identifiers* option, with the organism restricted to *Homo sapiens*.

The STRING network was constructed using the *full STRING network* option, integrating both functional and physical protein associations. All evidence channels were enabled (text mining, experiments, curated databases, co-expression, neighborhood, gene fusion, and co-occurrence), and the minimum required interaction score was set to *medium confidence* (0.400). First-shell interactors were included to obtain a comprehensive interaction network.

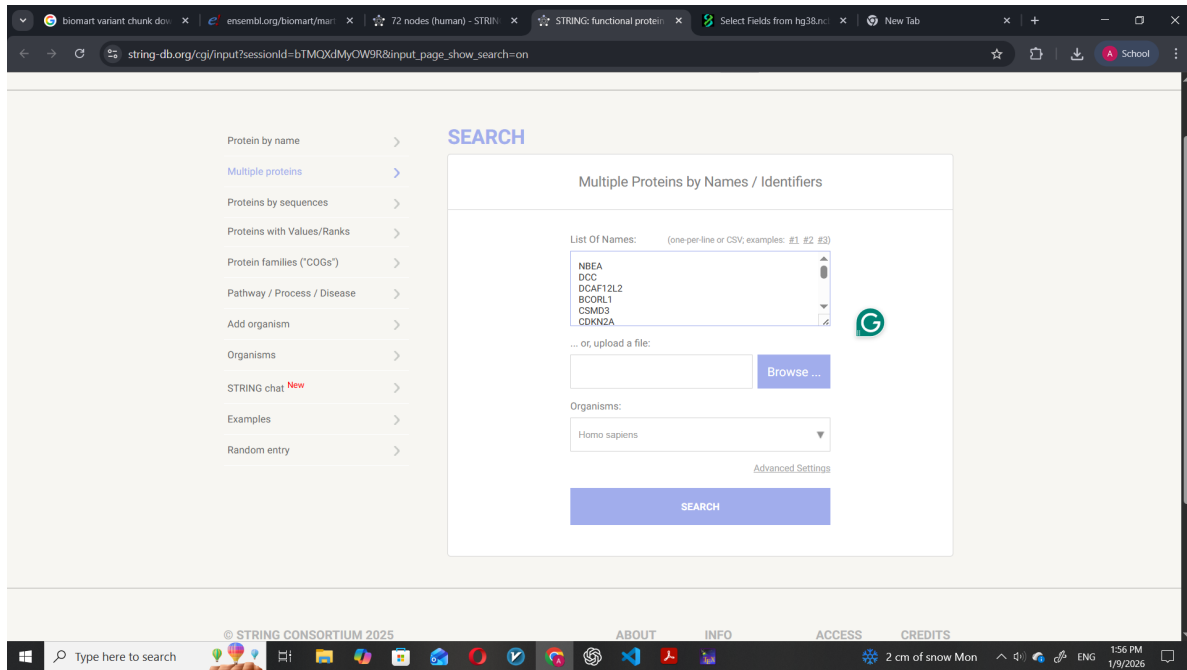


Figure 5. STRING multiple gene input from the results of the previous step. HGNC symbols are used in this step.

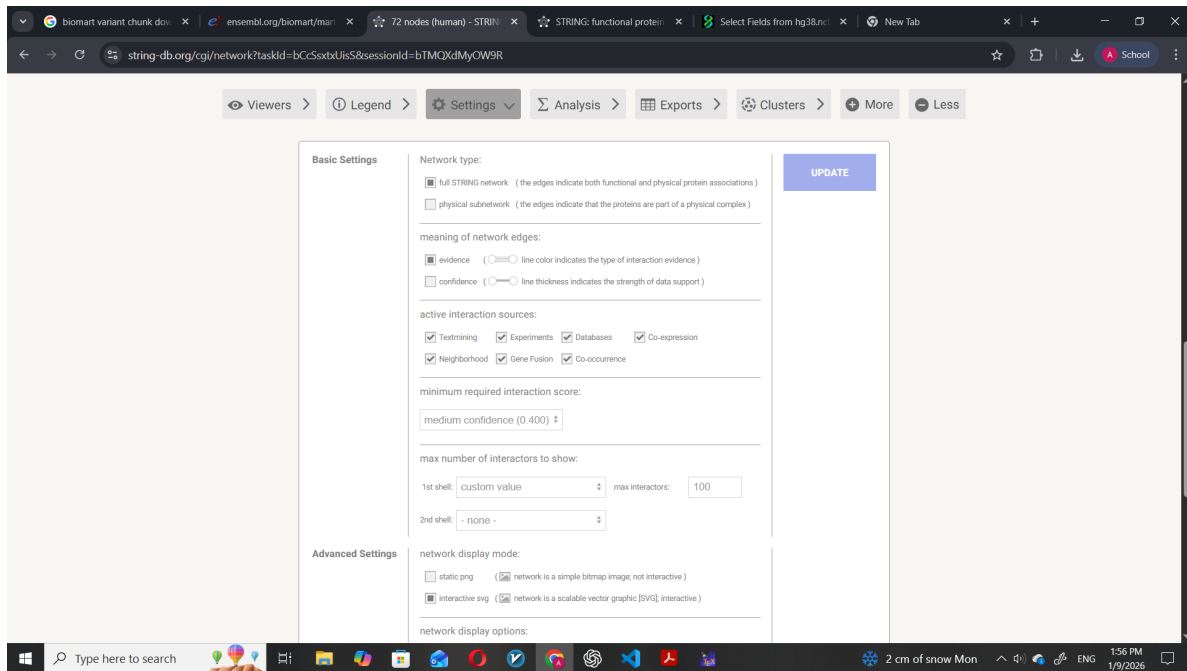


Figure 6. STRING setting max interaction to 100 according to the project description.

2.3 Step 3: Variant Retrieval Using Ensembl BioMart

The expanded gene set obtained from STRING was converted into Ensembl Gene Stable IDs to ensure compatibility with downstream genomic queries. These Ensembl IDs were then used to

retrieve genetic variants from the Ensembl BioMart `hsapiens_snp` dataset.

Variants were restricted to those originating from the dbSNP database by applying the `variation_source = dbSNP` filter. For each gene, the following variant-related attributes were retrieved: Ensembl Gene Stable ID, gene name, dbSNP identifier, chromosome name, variant start and end positions, alleles, and UCSC genome browser identifier (when available).

Due to instability and frequent timeouts when querying BioMart for large gene lists, a custom Python script was implemented to automate the querying process. The script submits XML-based BioMart queries programmatically and cycles through multiple Ensembl mirror servers to maximize query reliability.

To avoid server-side failures, the gene list was divided into small chunks, with a final stable configuration of four genes per query (`CHUNK_SIZE = 4`). Each chunk was queried sequentially, and results were appended to a unified output file. The complete variant retrieval process required approximately 20 hours to finish.

Screenshots of the BioMart configuration for variant-related attributes are shown in Figure 7.

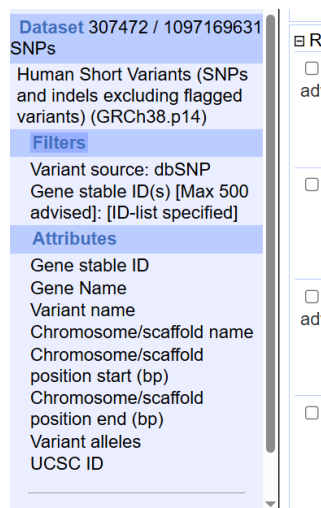


Figure 7. BioMart interface showing variant-related attribute selection for dbSNP queries.

2.4 Automated Variant Retrieval Script (Final Working Version)

Because BioMart queries for variant-level data frequently failed through the web interface (timeouts, incomplete output, or session termination), we implemented a Python script that programmatically submits BioMart XML queries to Ensembl’s `martservice` endpoint. The script automates three key requirements: (1) chunking the input gene list to avoid request size/time limits, (2) retrying failed requests and rotating across multiple Ensembl mirror servers, and (3) storing each chunk result independently, then merging all chunk files into a final unified TSV output.

Procedure (end-to-end). The pipeline implemented by the script is:

1. Read the list of Ensembl Gene Stable IDs (`ENSG...`) produced from the STRING-derived gene set.

2. Split genes into fixed-size chunks (`CHUNK_SIZE=4`) and build a BioMart XML query for each chunk.
3. Submit each query to BioMart using HTTP POST; on failures, retry and switch between Ensembl mirrors.
4. Save each chunk response to its own TSV file in `chunks/`.
5. Merge all chunk TSV files into a final output file while keeping a single header line.
6. Track which gene IDs actually appear in returned rows and write the remaining IDs into `missing_gene_ids.txt`.

Functionality of major methods.

- `read_gene_ids()`: reads the input file and extracts only valid Ensembl gene IDs while removing duplicates.
- `build_query_xml()`: constructs the XML query specifying dataset (`hsapiens_snp`), filters (dbSNP source and gene list), and output attributes (gene ID/name, rsID, coordinates, alleles, UCSC ID).
- `post_query()`: performs an HTTP POST to a BioMart endpoint with a timeout; raises an exception on HTTP failure.
- `query_with_mirrors()`: tries multiple Ensembl mirrors and retries each mirror several times before declaring the chunk failed.
- `returned_gene_ids_from_tsv()`: parses returned TSV rows and collects gene IDs that appear in the first column, enabling accurate “missing gene” detection even when BioMart returns partial results.
- `write_chunk_file()`: writes each chunk response into a separate file to prevent losing progress after long runs.
- `merge_chunk_files()`: merges chunk outputs into a single TSV, ensuring only one header is kept even though every chunk is requested with a header.
- `run()`: orchestrates the entire process, cleans old chunk outputs, writes logs, and produces both final data output and missing-gene list.

Final code used in the project. The following script is the final working version used to retrieve all dbSNP variants for the STRING-expanded schwannoma-associated gene set. In our final run, `CHUNK_SIZE=4` produced stable requests, but the full download still required approximately 20 hours due to BioMart rate limits and intermittent failures.

Listing 1. Final working BioMart variant retrieval script (chunked, mirror-aware, merge-based).

```
#!/usr/bin/env python3
from __future__ import annotations

import sys
import time
from typing import List, Tuple
import requests

# =====
# DEBUG DEFAULTS (NO TERMINAL INPUT NEEDED)
# Edit these paths for your machine if needed.
# =====

GENE_FILE = r"string_proteins_gene_stable_id_full.txt"
OUT_TSV = r"biomart_variants.tsv"
OUT_MISS = r"missing_gene_ids.txt"

DATASET = "hsapiens_snp"
CHUNK_SIZE = 4

MIRRORS = [
    "https://www.ensembl.org/biomart/martservice",
    "https://uswest.ensembl.org/biomart/martservice",
    "https://useast.ensembl.org/biomart/martservice",
    "https://asia.ensembl.org/biomart/martservice",
]

# From your BioMart XML view:
FILTER_SOURCE_NAME = "variation_source"
FILTER_GENE_NAME = "ensembl_gene"
FILTER_SOURCE_VALUE = "dbSNP"

ATTRIBUTES = [
    "ensembl_gene_stable_id",
    "ensembl_gene_name",
    "refsnp_id",
    "chr_name",
    "chrom_start",
    "chrom_end",
    "allele",
    "ucsc_id",
]

LOG_FILE = "run.log"

def returned_gene_ids_from_tsv(tsv_text: str) -> set:
```

```

out = set()
for ln in tsv_text.splitlines():
    if not ln.strip():
        continue
    if ln.startswith("Gene stable ID"): # skip header if present
        continue
    out.add(ln.split("\t", 1)[0]) # first column is gene stable id
return out

def log(msg: str) -> None:
    print(msg)
    with open(LOG_FILE, "a", encoding="utf-8", newline="\n") as f:
        f.write(msg + "\n")

def read_gene_ids(path: str) -> List[str]:
    ids = []
    with open(path, "r", encoding="utf-8") as f:
        for line in f:
            s = line.strip()
            if not s:
                continue
            if s.lower().startswith("gene") or s.lower().startswith("ensembl"):
                continue
            if s.startswith("ENSG"):
                ids.append(s)

    seen = set()
    out = []
    for g in ids:
        if g not in seen:
            seen.add(g)
            out.append(g)
    return out

def build_query_xml(dataset: str, genes: List[str], header: int = 1) -> str:
    gene_list = ",".join(genes)
    attrs = "\n".join([f'    <Attribute name = "{a}" />' for a in ATTRIBUTES])
    return f'<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE Query>
<Query virtualSchemaName = "default" formatter = "TSV" header = "{header}"
uniqueRows = "1" count = "" datasetConfigVersion = "0.6" >
    <Dataset name = "{dataset}" interface = "default" >
    <Filter name = "{FILTER_SOURCE_NAME}" value = "{FILTER_SOURCE_VALUE}" />
    <Filter name = "{FILTER_GENE_NAME}" value = "{gene_list}" />
    {attrs}
    </Dataset>
</Query>'

```



```

"""

def post_query(mirror: str, xml: str, timeout: int = 180) -> str:
    r = requests.post(mirror, data={"query": xml}, timeout=timeout)
    r.raise_for_status()
    return r.text

def is_biomart_error(txt: str) -> bool:
    low = txt.lower()
    return ("query error" in low) or ("biomart::exception" in low) or ("dataset"
in low and "not found" in low)

def query_with_mirrors(xml: str, retries: int = 3) -> Tuple[str, str]:
    last_err = None
    for mirror in MIRRORS:
        for attempt in range(1, retries + 1):
            try:
                txt = post_query(mirror, xml)
                if is_biomart_error(txt):
                    raise RuntimeError(txt.splitlines()[0] if txt else "BioMart
error")

                return txt, mirror
            except Exception as e:
                last_err = e
                time.sleep(1.2 * attempt)
    raise RuntimeError(f"All mirrors failed. Last error: {last_err}")

def parse_returned_gene_ids(tsv: str) -> set:
    lines = [ln for ln in tsv.splitlines() if ln.strip()]
    if len(lines) < 2:
        return set()
    header = lines[0].split("\t")
    idx = None
    for key in ("Ensembl Gene Stable ID", "ensembl_gene_stable_id"):
        if key in header:
            idx = header.index(key)
            break
    if idx is None:
        return set()
    out = set()
    for ln in lines[1:]:
        parts = ln.split("\t")
        if len(parts) > idx:
            out.add(parts[idx])
    return out

import os

```

```

from pathlib import Path

def write_chunk_file(chunk_dir: str, chunk_index: int, txt: str) -> str:
    """
    Writes one chunk output to a separate TSV file.
    Returns the filepath.
    """
    Path(chunk_dir).mkdir(parents=True, exist_ok=True)
    fp = os.path.join(chunk_dir, f"biomart_variants_chunk_{chunk_index:04d}.tsv")
    with open(fp, "w", encoding="utf-8", newline="\n") as f:
        f.write(txt if txt.endswith("\n") else (txt + "\n"))
    return fp

def merge_chunk_files(chunk_files: List[str], out_tsv: str) -> None:
    """
    Merges chunk TSV files into a single TSV.
    Keeps exactly ONE header (from the first chunk file that has a header).
    If chunks have no header (header=0), it merges as-is.
    """
    # Determine if the first non-empty chunk begins with a header line
    header_line = None
    first_data_seen = False

    with open(out_tsv, "w", encoding="utf-8", newline="\n") as out:
        for fp in chunk_files:
            with open(fp, "r", encoding="utf-8") as f:
                lines = f.read().splitlines()

                # Skip completely empty chunk files
                if not any(ln.strip() for ln in lines):
                    continue

                # If we haven't decided on a header yet, detect it
                if header_line is None:
                    first_nonempty = next((ln for ln in lines if ln.strip()), "")
                    if first_nonempty.startswith("Gene stable ID"):
                        header_line = first_nonempty
                        out.write(header_line + "\n")
                        # write remaining lines excluding header
                        for ln in lines[1:]:
                            if ln.strip():
                                out.write(ln + "\n")
                        first_data_seen = True
                        continue
                else:
                    # No header in this file; just write all lines
                    for ln in lines:

```

```

        if ln.strip():
            out.write(ln + "\n")
        first_data_seen = True
        continue

    # Header already known:
    if lines and lines[0].startswith("Gene stable ID"):
        # Skip the header line in subsequent chunks
        for ln in lines[1:]:
            if ln.strip():
                out.write(ln + "\n")
    else:
        for ln in lines:
            if ln.strip():
                out.write(ln + "\n")

    if not first_data_seen:
        # Ensure output exists but is empty if nothing was merged
        open(out_tsv, "w", encoding="utf-8").close()

def run():
    import os

    # reset log
    open(LOG_FILE, "w", encoding="utf-8").close()

    log(f"Using dataset={DATASET}")
    log(f"Gene file: {GENE_FILE}")
    log(f"Output TSV: {OUT_TSV}")
    log(f"Chunk size: {CHUNK_SIZE}")

    gene_ids = read_gene_ids(GENE_FILE)
    if not gene_ids:
        log("ERROR: No ENSG IDs found in input file.")
        sys.exit(1)

    log(f"Input genes: {len(gene_ids)}")
    log(f"First 3 gene IDs: {gene_ids[:3]}")

    # Directory to store chunk results
    chunk_dir = "chunks"

    # Clean old chunk files (avoid mixing old + new)
    if os.path.isdir(chunk_dir):
        for fn in os.listdir(chunk_dir):
            if fn.startswith("biomart_variants_chunk_") and fn.endswith(".tsv"):
                try:

```

```

        os.remove(os.path.join(chunk_dir, fn))
    except Exception:
        pass
else:
    os.makedirs(chunk_dir, exist_ok=True)

returned_genes_all = set()
chunk_files: List[str] = []
failed_chunks: List[int] = []

# We will request header=1 for EVERY chunk file.
# Reason: simplifies merge + missing gene detection.
# We will remove duplicate headers at merge time.
for idx, i in enumerate(range(0, len(gene_ids), CHUNK_SIZE), start=1):
    chunk_genes = gene_ids[i:i + CHUNK_SIZE]
    xml = build_query_xml(DATASET, chunk_genes, header=1)

    # keep last query for debugging
    with open("last_query.xml", "w", encoding="utf-8") as f:
        f.write(xml)

    try:
        txt, used = query_with_mirrors(xml)
        log(f"Chunk {idx}: genes {i+1}-{min(i+CHUNK_SIZE, len(gene_ids))} OK (
mirror: {used})")
        log(f"Response length: {len(txt)} chars")

        # Save this chunk output separately
        fp = write_chunk_file(chunk_dir, idx, txt)
        chunk_files.append(fp)

        # Update returned genes set from the real returned TSV text
        returned_genes_all |= returned_gene_ids_from_tsv(txt)

    except Exception as e:
        failed_chunks.append(idx)
        log(f"ERROR: Chunk {idx} failed. Reason: {e}")

    time.sleep(0.4)

# Merge chunk files into final output
if chunk_files:
    merge_chunk_files(chunk_files, OUT_TSV)
    try:
        size = __import__("os").path.getsize(OUT_TSV)
        log(f"Final merged TSV size: {size} bytes")
    except Exception as e:

```

```

        log(f"Could not stat merged output TSV: {e}")
    else:
        # no chunks succeeded
        open(OUT_TSV, "w", encoding="utf-8").close()
        log("WARNING: No chunk files produced. Output TSV is empty.")

        # Missing genes = those never observed in returned data
        missing = [g for g in gene_ids if g not in returned_genes_all]
        with open(OUT_MISS, "w", encoding="utf-8", newline="\n") as f:
            f.write("\n".join(missing) + ("\n" if missing else ""))

        log(f"Genes with no returned rows: {len(missing)} -> {OUT_MISS}")
        if failed_chunks:
            log(f"Failed chunks: {failed_chunks} (you can re-run; successful chunks are
preserved in {chunk_dir}/)")
        log("DONE.")

    if __name__ == "__main__":
        run()

```

3 Part 3: Genomic Coordinates and Transcript Annotation

In this part, genomic coordinates and transcript-level structural information were retrieved for the genes identified in Part 2. Since the STRING database gene list was converted to Ensembl Gene IDs, and UCSC RefSeq tables are transcript-centric, we adopted a transcript-level representation in order to preserve all available annotation without introducing arbitrary transcript selection criteria.

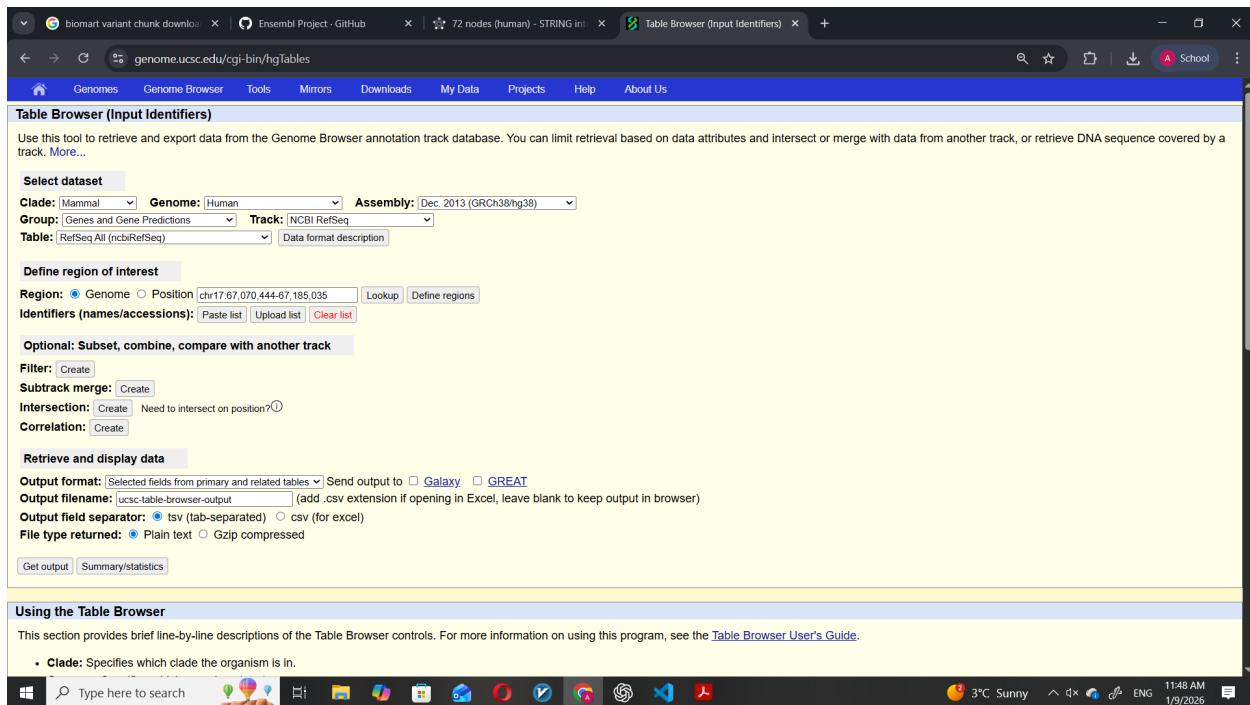


Figure 8. UCSC Table Browser configuration showing genome assembly (hg38), RefSeq track, and ncbiRefSeq table selection.

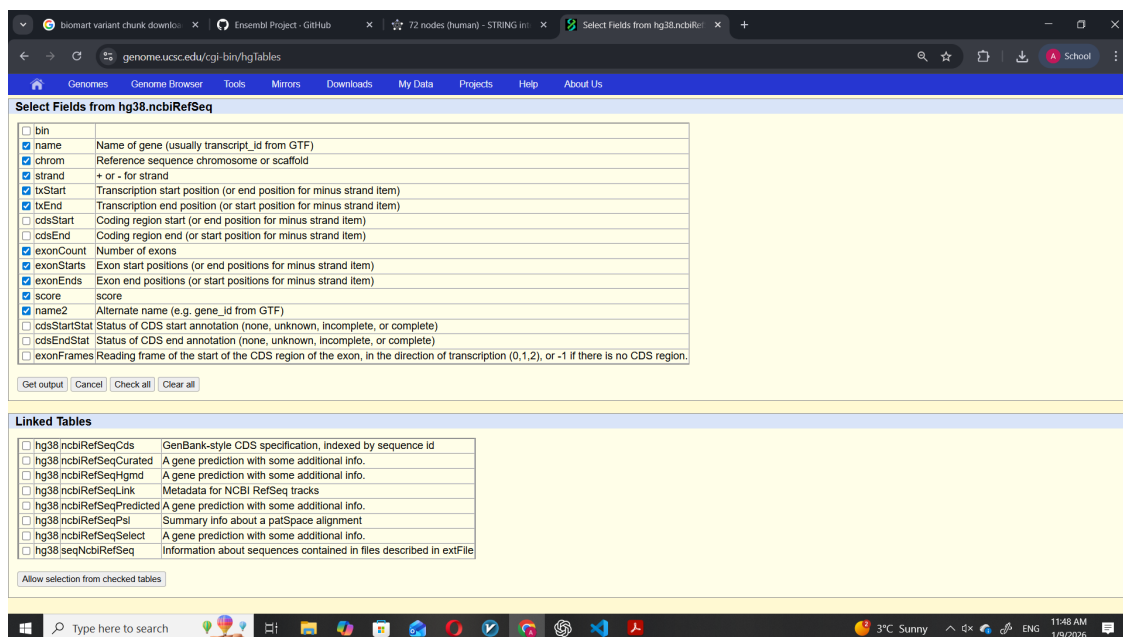


Figure 9. Selection of transcript-level fields from the hg38.ncbiRefSeq table, including genomic coordinates and exon structure.

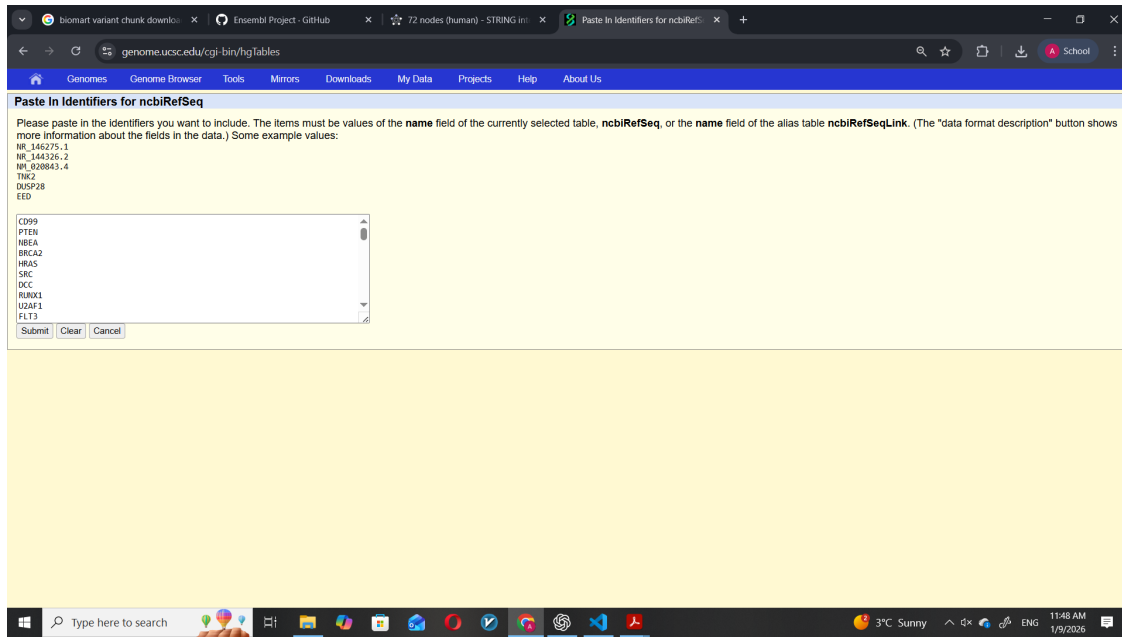


Figure 10. Input of gene symbols into the UCSC Table Browser identifier field for RefSeq-based transcript retrieval.

3.1 Data Source and Genome Assembly

Genomic annotations were obtained using the **UCSC Table Browser**. All queries were performed on the human genome assembly **GRCh38 (hg38)**. The following configuration was used consistently throughout this step:

- Clade: Mammal
- Genome: Human
- Assembly: GRCh38 (hg38)
- Group: Genes and Gene Predictions
- Track: NCBI RefSeq
- Table: ncbiRefSeq

Figure 8 shows the Table Browser configuration used to select the appropriate dataset and genome assembly.

3.2 Input Identifiers

Gene symbols corresponding to the Ensembl Gene IDs obtained in Part 2 were used as input identifiers. These symbols were pasted directly into the *Identifiers (names/accessions)* field of the UCSC Table Browser. The **ncbiRefSeq** table accepts gene symbols and RefSeq transcript accessions through its primary and alias tables, enabling successful mapping for the majority of genes.

An example of the identifier input interface is shown in Figure 10.

3.3 Selected Attributes

To comprehensively capture transcript-level genomic structure, the following fields were selected from the `hg38.ncbiRefSeq` table:

- `name` (RefSeq transcript accession)
- `chrom` (chromosome)
- `strand`
- `txStart`, `txEnd` (transcription start and end positions)
- `exonCount`
- `exonStarts`
- `exonEnds`
- `name2` (gene symbol)

These fields provide both gene-level identifiers and detailed exon–intron structure for each transcript. Figure 9 illustrates the field selection interface used in this step.

3.4 Transcript-Level Representation

The UCSC RefSeq table returns multiple transcript entries for many genes, including protein-coding (e.g., `NM_`, `XM_`) and non-coding (e.g., `NR_`, `XR_`) transcripts. Rather than selecting a single representative transcript per gene, all returned transcripts were retained.

This design choice ensures:

- no loss of structural information,
- compatibility with alternative splicing analysis,
- transparent linkage between gene symbols and multiple transcript isoforms.

Each transcript is uniquely identified by its RefSeq accession and linked to the corresponding gene symbol via the `name2` field.

3.5 Output Format and Downstream Use

The resulting data were exported in tab-separated (TSV) format and later imported into a relational database. Each row corresponds to a single RefSeq transcript, with exon boundaries stored as comma-separated coordinate lists.

This transcript-level table serves as the genomic backbone for subsequent project components, enabling joins with variant data (Part 2) and protein-level annotations in later stages of the project.

4 Part 4 — UniProt ID Mapping (Genes → Proteins)

I got the Ensembl Gene IDs from the previous part and used the UniProt website to map them to UniProt IDs.

Retrieve/ID mappingⁱ

Enter one or more IDs (100,000 max). You may also [load from a text file](#). Separate IDs by whitespace (space, tab, newline) or commas. This service can also be used [programmatically](#). Alternatively, the [underlying data](#) can be downloaded.

ENSG00000051341
ENSG00000051382
ENSG00000057657
ENSG00000066032
ENSG00000066279
ENSG00000066468
ENSG00000070756
ENSG00000076685
ENSG00000081237
ENSG00000083857
ENSG00000085185

! Your input contains 72 unique IDs

From database To database

UniProtKB AC/ID UniProtKB

Name your ID Mapping job

ENSG00000051341 +71 UniProtKB_AC-ID → UniProtKB

Reset Map 72 IDs

Figure 11. ID mapping interface showing the mapping of Ensembl Gene IDs to UniProt IDs.

Then I downloaded the data and saved it in a ‘idmapping.tsv’ TSV file. Fortunately, there were not any missing genome that has not protein. and as mentioned in the documentation the columns are added.

5 Part 5 — GEO Dataset & GEO2R

Describe GEO search constraints (human, since Jan 1 2005), chosen dataset, and the GEO2R output table.

6 Part 6 — Additional Table from Another Database

I got the AlphaFoldDB IDs from the previous part and used the CATH database to map them to CATH domains.

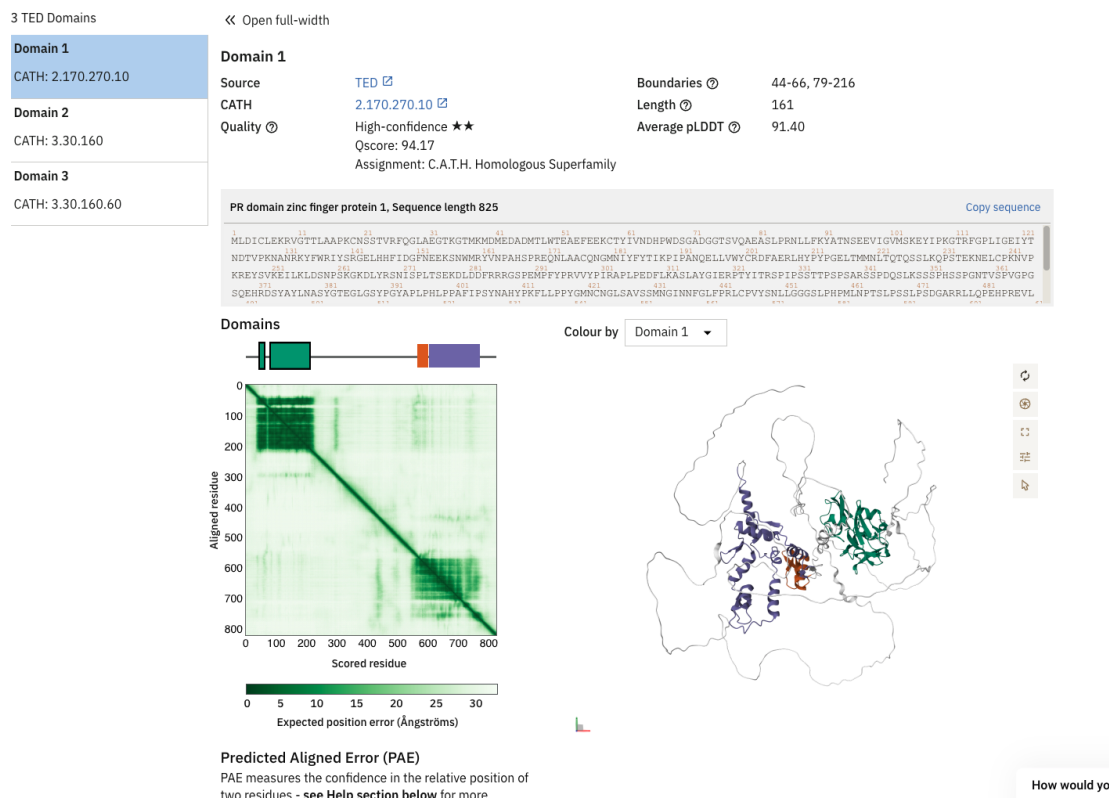


Figure 12. AlphaFoldDB CATH mapping interface showing the mapping of AlphaFoldDB IDs to CATH domains.

Then I downloaded the data and saved it in a `alphafoldcath.csv` CSV file.

6.1 Integration of TED and AlphaFold Database

Recent advances in protein structure prediction, particularly through AlphaFold, have enabled the availability of millions of predicted protein structures. However, large-scale structural availability alone is not sufficient for understanding protein function, evolution, and modular organization. To address this, domain-level annotation frameworks such as the **Truncated Evolutionary Domain (TED)** database have been developed and integrated with structural classification systems like **CATH**, leveraging the AlphaFold Database (AFDB) as a structural backbone.

6.2 The Truncated Evolutionary Domain (TED) Database

TED is a domain-centric resource designed to identify and annotate *evolutionary protein domains* directly from predicted protein structures. Unlike traditional domain databases that rely heavily on sequence similarity, TED operates primarily at the structural level, enabling the discovery of both known and previously uncharacterized domains. A key feature of TED is its ability to segment full-length protein structures into compact, evolutionarily meaningful domains, even in cases where sequence-based methods fail. This is particularly important for AlphaFold-predicted structures, which often contain long, multi-domain proteins with low sequence homology to known proteins.

TED therefore enables large-scale, structure-driven domain discovery across proteomes.

6.3 CATH Structural Classification

CATH is a hierarchical protein domain classification system that organizes protein domains based on structural and evolutionary relationships. It categorizes domains into four principal levels: *Class*, *Architecture*, *Topology*, and *Homologous superfamily*. Each level captures increasingly specific structural and evolutionary features. CATH serves as a gold-standard reference for known protein domain families. However, its coverage has historically been limited by the availability of experimentally determined structures. The emergence of high-quality predicted structures from AlphaFold has significantly expanded the potential domain space that can be analyzed and classified.

6.4 Integration with AlphaFold Database (AFDB)

The AlphaFold Database provides predicted three-dimensional structures for entire proteomes at unprecedented scale. TED leverages AFDB by using AlphaFold-generated models as input, performing automated domain boundary detection on these structures. The resulting TED domains are then structurally compared against existing CATH domains. If a TED-identified domain matches a known CATH homologous superfamily, it can be assigned accordingly, effectively extending CATH annotations into the predicted structure space. Domains that do not match existing CATH families are retained as novel evolutionary domains, highlighting potential new folds or previously unrecognized domain families. This integration creates a powerful pipeline in which AFDB supplies large-scale structural data, TED performs structure-based domain segmentation, and CATH provides evolutionary classification and contextualization. Together, they enable comprehensive, proteome-wide domain annotation that bridges the gap between structure prediction and functional/evolutionary interpretation.

6.5 Impact and Applications

The TED–CATH–AFDB integration substantially expands the known domain universe and improves the annotation of multi-domain proteins, particularly in poorly characterized organisms. This framework is especially valuable for downstream applications such as protein function prediction, evolutionary analysis, and structure-guided machine learning models that operate at the domain level rather than full-length sequences.

7 Part 7 — Conceptual Design (ER Diagram + Keys)

The conceptual design of the Schwannoma database integrates data from multiple biological databases to create a unified schema that links phenotype-related genes, genetic variants, genomic coordinates, protein annotations, protein interactions, structural domains, and gene expression data. The database follows a relational model with well-defined entities and relationships that preserve referential integrity across different data sources.

7.1 Entity-Relationship Diagram

The database schema consists of seven main entities (tables) with the following relationships:

- **Genes** (central entity): Contains core gene information from Ensembl/BioMart. Each gene is uniquely identified by its Ensembl Gene Stable ID.
- **Variants**: Stores genetic variants (SNPs and indels) from dbSNP associated with genes. Multiple variants can be associated with a single gene (one-to-many relationship: one gene to many variants).
- **Transcripts**: Contains RefSeq transcript-level genomic coordinates and exon structure from UCSC Table Browser. Multiple transcripts can be associated with a single gene symbol (one-to-many relationship: one gene to many transcripts).
- **Proteins**: Maps genes to their corresponding UniProt protein entries. Multiple proteins can be encoded by a single gene due to alternative splicing and isoform diversity (one-to-many relationship: one gene to many proteins).
- **STRING_Interactions**: Represents protein-protein interactions from the STRING database. This is a many-to-many relationship between genes/proteins, where each interaction links two gene symbols.
- **AlphaFold_CATH**: Contains structural domain annotations from AlphaFold and CATH databases, linked to proteins. Multiple structural domains (TEDs) can be associated with a single protein (one-to-many relationship: one protein to many domains).
- **GEO2R_Results**: Stores differential gene expression analysis results from GEO2R. Each row represents expression data for a gene symbol (one-to-one or one-to-many relationship: one gene to one or more expression records).

7.1.1 Cardinality Ratios

The cardinality ratios between entities are as follows:

- **Genes** → **Variants**: One-to-Many (1:N). A single gene can have multiple associated variants, but each variant is associated with exactly one gene.
- **Genes** → **Transcripts**: One-to-Many (1:N). A single gene (identified by gene symbol) can have multiple transcript isoforms, but each transcript belongs to exactly one gene.
- **Genes** → **Proteins**: One-to-Many (1:N). A single gene can encode multiple protein isoforms, but each protein is encoded by exactly one gene.
- **Proteins** → **AlphaFold_CATH**: One-to-Many (1:N). A single protein can contain multiple structural domains (TEDs), but each domain belongs to exactly one protein.

- **Genes $\leftrightarrow$ STRING_Interactions:** Many-to-Many (M:N). Genes can interact with multiple other genes, and each interaction involves exactly two genes. This relationship is represented through a composite key in the interactions table.
- **Genes $\rightarrow$ GEO2R_Results:** One-to-Many (1:N). A single gene can have multiple expression records (e.g., from different comparisons or time points), but each expression record corresponds to exactly one gene.

7.2 Primary Keys and Foreign Keys

The following table summarizes the primary keys (PK) and foreign keys (FK) for each relation in the database:

Table Name	Primary Key	Foreign Keys
Genes	gene_stable_id (Ensembl Gene Stable ID)	None (root entity)
Variants	variant_id (composite: gene_stable_id, refsnp_id)	gene_stable_id $\rightarrow$ Genes.gene_stable_id
Transcripts	refseq_accession (RefSeq transcript accession, e.g., NM_*, XM_*)	gene_symbol $\rightarrow$ Genes.hgnc_symbol
Proteins	uniprot_id (UniProt Entry ID)	ensembl_gene_id $\rightarrow$ Genes.gene_stable_id
STRING_Interactions	interaction_id (composite: node1, node2)	node1 $\rightarrow$ Genes.hgnc_symbol, node2 $\rightarrow$ Genes.hgnc_symbol
AlphaFold_CATH	ted_id (TED domain identifier)	uniprot_acc $\rightarrow$ Proteins.uniprot_id
GEO2R_Results	geo_result_id (auto-increment) or composite: gene_symbol, comparison_id	gene_symbol $\rightarrow$ Genes.hgnc_symbol

Table 1. Primary keys and foreign keys for each table in the Schwannoma database.

7.2.1 Detailed Key Definitions

Genes Table

- **Primary Key:** gene_stable_id (VARCHAR) — Ensembl Gene Stable ID (e.g., ENSG00000172915). This is the central identifier that links to all other tables.
- **Additional Unique Constraint:** hgnc_symbol (VARCHAR) — HGNC gene symbol, used for linking to transcripts and interactions.

Variants Table

- **Primary Key:** Composite key consisting of `gene_stable_id` (VARCHAR) and `refsnp_id` (VARCHAR, e.g., `rs123456`). This ensures uniqueness since the same variant can theoretically appear in multiple genes (though rare).
- **Foreign Key:** `gene_stable_id` references `Genes.gene_stable_id` (ON DELETE CASCADE, ON UPDATE CASCADE).

Transcripts Table

- **Primary Key:** `refseq_accession` (VARCHAR) — RefSeq transcript identifier (e.g., `NM_001328648.1`, `XM_047422550.1`).
- **Foreign Key:** `gene_symbol` (VARCHAR) references `Genes.hgnc_symbol` (ON DELETE SET NULL, ON UPDATE CASCADE). Note: This uses gene symbol rather than Ensembl ID because UCSC RefSeq data is primarily linked via gene symbols.

Proteins Table

- **Primary Key:** `uniprot_id` (VARCHAR) — UniProt Entry ID (e.g., `075417`, `P42338`).
- **Foreign Key:** `ensembl_gene_id` (VARCHAR) references `Genes.gene_stable_id` (ON DELETE CASCADE, ON UPDATE CASCADE).

STRING_Interactions Table

- **Primary Key:** Composite key consisting of `node1` (VARCHAR) and `node2` (VARCHAR), where both are HGNC gene symbols. To ensure consistency, we enforce `node1 < node2` lexicographically to avoid duplicate bidirectional entries.
- **Foreign Keys:**
 - `node1` references `Genes.hgnc_symbol` (ON DELETE CASCADE, ON UPDATE CASCADE)
 - `node2` references `Genes.hgnc_symbol` (ON DELETE CASCADE, ON UPDATE CASCADE)

AlphaFold_CATH Table

- **Primary Key:** `ted_id` (VARCHAR) — Truncated Evolutionary Domain identifier (e.g., `AF-075417-F1-model_v4-TED01`).
- **Foreign Key:** `uniprot_acc` (VARCHAR) references `Proteins.uniprot_id` (ON DELETE CASCADE, ON UPDATE CASCADE).

GEO2R_Results Table

- **Primary Key:** `geo_result_id` (INTEGER, auto-increment) or composite key of `gene_symbol` and `comparison_id` if multiple comparisons exist.
- **Foreign Key:** `gene_symbol` (VARCHAR) references `Genes.hgnc_symbol` (ON DELETE CASCADE, ON UPDATE CASCADE).
- **Additional Attributes:** Includes `logFC` (log fold change), `adj_p_value` (adjusted p-value), `p_value`, and other GEO2R analysis metrics.

7.2.2 Referential Integrity Constraints

To maintain data consistency, the following referential integrity constraints are enforced:

- **Cascade Deletes:** When a gene is deleted from the `Genes` table, all associated variants, proteins, and GEO2R results are automatically deleted. Interactions involving the deleted gene are also removed.
- **Cascade Updates:** If an Ensembl Gene Stable ID or HGNC symbol is updated, all foreign key references are automatically updated to maintain consistency.
- **Set NULL on Delete:** For the `Transcripts` table, if a gene symbol is deleted, the `gene_symbol` foreign key is set to NULL rather than deleting the transcript record, preserving transcript-level data even if gene mapping becomes unavailable.

7.2.3 Additional Constraints

Beyond primary and foreign keys, the following constraints ensure data quality:

- **CHECK Constraints:**
 - `Variants.chrom_start ≤ Variants.chrom_end`
 - `Transcripts.txStart ≤ Transcripts.txEnd`
 - `Transcripts.exonCount ≥ 1`
 - `Proteins.length > 0`
 - `GEO2R_Results.adj_p_value ≥ 0 AND ≤ 1`
- **UNIQUE Constraints:**
 - `Genes.hgnc_symbol` (if not NULL) must be unique
 - `Proteins.uniprot_id` must be unique
 - `Transcripts.refseq_accession` must be unique
- **NOT NULL Constraints:** Critical identifier fields such as `Genes.gene_stable_id`, `Variants.refsnp_id`, `Proteins.uniprot_id`, and `Transcripts.refseq_accession` are enforced as NOT NULL.

This conceptual design provides a robust foundation for integrating multi-omics data related to schwannoma, enabling complex queries that span genomic, transcriptomic, proteomic, and expression domains.

8 Challenges and Ambiguities

8.1 Code Evolution and Rationale (Old vs. New Implementation)

During Part 2, we iteratively improved the variant-retrieval code as repeated failures appeared when scaling from small test gene lists to the full STRING-expanded set. The core difficulties were BioMart instability (timeouts, truncated responses such as “Response ended prematurely”), long runtimes, and incorrect missing-gene reporting when the returned output was incomplete.

Old implementation (direct append approach). The earlier version of the script appended each chunk response directly into the final TSV file as the run progressed, and it tracked missing genes using a header-dependent parsing strategy that effectively only validated the first chunk. This older strategy was workable for very small gene lists but was fragile when the service intermittently failed.

Drawbacks of the old approach:

- **Progress was not safely checkpointed:** if a late chunk failed after many hours, there was no reliable way to resume without re-running everything.
- **Missing-gene detection was inaccurate:** the code only accumulated returned gene IDs from the first chunk (the only chunk guaranteed to include the header), causing many genes to be incorrectly written into `missing_gene_ids.txt` even when their variants existed in later chunks.
- **Hard to debug partial results:** all outputs were mixed into one file, so identifying which chunk caused corruption or truncation was difficult.

Listing 2. Old BioMart variant retrieval script (single output append, fragile missing-gene tracking).

```
def run():
    # reset log
    open(LOG_FILE, "w", encoding="utf-8").close()

    log(f"Using dataset={DATASET}")
    log(f"Gene file: {GENE_FILE}")
    log(f"Output TSV: {OUT_TSV}")
    log(f"Chunk size: {CHUNK_SIZE}")

    gene_ids = read_gene_ids(GENE_FILE)
    if not gene_ids:
        log("ERROR: No ENSG IDs found in input file.")
```



```

        sys.exit(1)

    log(f"Input genes: {len(gene_ids)}")
    log(f"First 3 gene IDs: {gene_ids[:3]}")

    # reset outputs
    open(OUT_TSV, "w", encoding="utf-8").close()

    header_written = False
    returned_genes_all = set()

    for i in range(0, len(gene_ids), CHUNK_SIZE):
        chunk_genes = gene_ids[i:i + CHUNK_SIZE]

        # header=1 only for first chunk; then header=0
        xml = build_query_xml(DATASET, chunk_genes, header=1 if not
header_written else 0)

        # For debugging: write the XML being sent
        with open("last_query.xml", "w", encoding="utf-8") as f:
            f.write(xml)

        txt, used = query_with_mirrors(xml)
        log(f"Chunk {i//CHUNK_SIZE + 1}: genes {i+1}-{min(i+CHUNK_SIZE, len(
gene_ids))} OK (mirror: {used})")
        log(f"Response length: {len(txt)} chars")

        with open(OUT_TSV, "a", encoding="utf-8", newline="\n") as f:
            f.write(txt if txt.endswith("\n") else (txt + "\n"))

        # track returned genes from first chunk only (has header)
        if not header_written:
            returned_genes_all |= parse_returned_gene_ids(txt)
            header_written = True

        time.sleep(0.4)

    # If TSV is empty, log it explicitly
    try:
        size = __import__("os").path.getsize(OUT_TSV)
        log(f"Final TSV size: {size} bytes")
        if size < 5:
            log("WARNING: TSV file is essentially empty. Check last_query.xml and
run.log.")
    except Exception as e:
        log(f"Could not stat output TSV: {e}")

```

```

missing = [g for g in gene_ids if g not in returned_genes_all]
with open(OUT_MISS, "w", encoding="utf-8", newline="\n") as f:
    f.write("\n".join(missing) + ("\n" if missing else ""))

log(f"Genes not observed in first-chunk tracking: {len(missing)} -> {OUT_MISS}")

log("DONE.")

```

New implementation (chunk files + merge + accurate tracking). To address these issues, we redesigned the script around **checkpointed chunk files**. Each chunk is written into a separate TSV file under `chunks/`. This makes the pipeline resilient: if BioMart fails mid-run, successful chunks remain saved and can be reused without recomputing earlier results. Additionally, the new implementation requests a header for every chunk (`header=1`) and removes duplicate headers during the merge step. This greatly simplifies debugging and allows consistent parsing.

The “missing genes” list is now computed from the actual returned rows across *all* chunk outputs using `returned_gene_ids_from_tsv()`, which does not rely on the presence of a header or a specific column index.

Advantages of the new approach:

- **Checkpointing:** intermediate results persist even if the run takes many hours.
- **Correct missing-gene reporting:** genes are marked missing only if they never appear in any returned row across all chunks.
- **Traceability:** errors can be linked to a specific chunk file and re-run strategy can focus only on failed indices.
- **Stable merging:** output is normalized into one TSV with a single header.

This final version (Section 2.4) successfully completed the full variant download with `CHUNK_SIZE=4`, producing a comprehensive variant dataset, although the total runtime was approximately 20 hours due to service constraints and enforced throttling.

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